Merging Linked Lists

AlgoExpert · Medium · Linked Lists

The problem

- Two lists may converge into a shared tail; return the first shared node or null.
- Two pointers that switch lists meet at the junction after at most n+m steps.

Intuition

- Two pointers that switch to the other list after finishing travel the same distance and meet at the junction.

Approach

- 1. Advance both pointers
- 2. When one ends, jump to the other list's head
- 3. They meet at the shared node or both null

Java solution

```java
public class MergingLinkedLists {
    public static LinkedListNode mergingLinkedLists(LinkedListNode a, LinkedListNode b) {
        LinkedListNode p = a, q = b;
        while (p != q) { p = p == null ? b : p.next; q = q == null ? a : q.next; }
        return p;
    }
}
```

Complexity

- Time: O(n + m)
- Space: O(1)

Recap

- Two pointers that switch to the other list after finishing travel the same distance and meet at the junction.
- Complexity: O(n + m)